

A commuting-symbol characterization for semicommutative martingale paraproducts

Automated mathematics run `fa_banach_001`, lane 8

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Status. Partial result, likely valid, subject to human review. It settles a natural commuting-symbol subcase of the open characterization problem, not the general noncommutative problem.

1 Source question

Wei and Zhang ask how to characterize boundedness of an operator-valued martingale paraproduct on semicommutative L_2 [1, p. 2]:

“It still remains open how to characterize the boundedness of π_b on $L_2(L_\infty(\mathbb{R})\overline{\otimes}\mathcal{M})$.”

The surrounding source passage, including the finite-dimensional logarithmic obstruction and the definition of the symbol class, is reproduced below.

Nazarov, Treil and Volberg also independently obtained (1.1) in [16] by the Bellman method, and they also gave an example showing that for any $n \in \mathbb{N}$ there exists b such that

$$\|\pi_b\|_{B(L_2(L_\infty(\mathbb{R})\overline{\otimes}\mathbb{M}_n))} \gtrsim \sqrt{\log(n+1)}\|b\|_{BMO_{so}^2(\mathbb{R},\mathbb{M}_n)}.$$

This implies that the boundedness of π_b cannot be characterized solely by $BMO_{so}^2(\mathbb{R},\mathcal{M})$ for infinite-dimensional $\mathcal{M}$. In [15], Nazarov, Pisier, Treil and Volberg proved that $\log(n+1)$ is the optimal order of the best constant in (1.1). It still remains open how to characterize the boundedness of π_b on $L_2(L_\infty(\mathbb{R})\overline{\otimes}\mathcal{M})$.

Although we do not know how to describe the boundedness of semicommutative martingale paraproducts until now, some particular properties of π_b can be investigated. For instance, Mei succeeded in describing the extrapolation property by virtue of $BMO_{\mathcal{M}}^2(\mathbb{R})$ for dyadic martingale paraproducts in [13]. Recall that $BMO_{\mathcal{M}}^d(\mathbb{R})$ is the operator-valued BMO space associated with the d -adic martingales consisting of all $\mathcal{M}$ -valued functions b that are Bochner integrable on any d -adic interval such that

$$(1.2) \quad \|b\|_{BMO_{\mathcal{M}}^d(\mathbb{R})} = \sup_{I \in \mathcal{D}} \left(\frac{1}{m(I)} \int_I \|b - (\frac{1}{m(I)} \int_I b \, dm)\|_{\mathcal{M}}^2 \, dm \right)^{1/2} < \infty.$$

Let $d \geq 2$, let $\mathcal{D}$ be the d -adic filtration on $\mathbb{R}$, and write $\mathbb{E}_k$ and $d_k = \mathbb{E}_k - \mathbb{E}_{k-1}$ for conditional expectation and martingale difference. If $(\mathcal{M}, \tau)$ is a semifinite von Neumann algebra, the semicommutative paraproduct is initially defined on finite martingales by

$$\pi_b f = \sum_k d_k b \mathbb{E}_{k-1} f.$$

We use the column martingale BMO seminorm

$$\|b\|_{BMO_{c,d}} := \sup_n \left\| \mathbb{E}_n \sum_{k \geq n} |d_k b|^2 \right\|_{\mathcal{M}}^{1/2}. \quad (1)$$

Changing $k \geq n$ to $k > n$ gives an equivalent seminorm, with constants depending only on d ; this elementary indexing point is included in the proof below.

2 Idea of the proof

The obstruction in the general problem is noncommutation of the operator coefficients. If all values of b lie in one abelian von Neumann algebra, left multiplication by those values is a commuting normal family on $L_2(\mathcal{M})$. The abelian spectral theorem simultaneously diagonalizes that family. Each finite truncation of π_b then becomes a direct integral of scalar paraproducts, tensored with identity operators on multiplicity spaces.

The operator norm of this direct integral is the essential supremum of its scalar fiber norms. Positivity and the same diagonalization show that (1) is precisely the essential supremum of the scalar fiber BMO seminorms. The scalar Carleson embedding theorem compares the scalar paraproduct and BMO norms. Uniform bounds for all truncations give the full operator. Thus the ambient algebra and the input may remain noncommutative; only the range of the symbol is required to commute.

3 Formal result

Theorem 1 (Commuting-range characterization). *Let $(\mathcal{M}, \tau)$ be a semifinite von Neumann algebra, and let $b : \mathbb{R} \rightarrow \mathcal{M}$ be locally Bochner integrable and adapted to the d -adic filtration. Suppose there is an abelian von Neumann subalgebra $\mathcal{A} \subseteq \mathcal{M}$ such that $b(t) \in \mathcal{A}$ for almost every t .*

Then π_b extends boundedly to

$$L_2(L_\infty(\mathbb{R}) \overline{\otimes} \mathcal{M}) \cong L_2(\mathbb{R}; L_2(\mathcal{M}, \tau))$$

if and only if $\|b\|_{\text{BMO}_{c,d}} < \infty$. Moreover, there are constants $0 < c_d \leq C_d < \infty$, depending only on d , such that

$$c_d \|b\|_{\text{BMO}_{c,d}} \leq \|\pi_b\|_{2 \rightarrow 2} \leq C_d \|b\|_{\text{BMO}_{c,d}}. \quad (2)$$

Corollary 2. *The conclusion holds whenever the operators in the essential range of b are normal and commute pairwise together with their adjoints. In particular, it holds for matrix-valued symbols which are simultaneously unitarily diagonalizable, with constants independent of the matrix size.*

4 Proof

We first record the scalar input in the precise form needed here.

Lemma 3 (Scalar paraproduct–Carleson equivalence). *For a scalar d -adic martingale β , let*

$$C(\beta)^2 = \sup_n \left\| \mathbb{E}_n \sum_{k>n} |d_k \beta|^2 \right\|_\infty.$$

Then the scalar paraproduct π_β on $L_2(\mathbb{R})$ is bounded if and only if $C(\beta) < \infty$, and

$$C(\beta) \leq \|\pi_\beta\|_{2 \rightarrow 2} \leq 2C(\beta).$$

Furthermore,

$$C(\beta) \asymp_d \sup_n \left\| \mathbb{E}_n \sum_{k \geq n} |d_k \beta|^2 \right\|_\infty^{1/2}.$$

The same estimates hold for Hilbert-space-valued inputs, since the scalar paraproduct is tensored with the identity.

Proof. The quantity $C(\beta)^2$ is the scalar martingale Carleson constant of the positive sequence $a_k = |d_k \beta|^2$. The usual stopping-time proof of the Carleson embedding theorem gives

$$\sum_k \int a_k |\mathbb{E}_{k-1} f|^2 dm \leq 4C(\beta)^2 \|f\|_2^2.$$

The left side equals $\|\pi_\beta f\|_2^2$, because the summands $d_k \beta \mathbb{E}_{k-1} f$ are orthogonal martingale differences. This proves the upper estimate. Conversely, test on indicators of finite-measure d -adic sets at level n . For all later levels the conditional expectation of such an indicator is the indicator itself, so boundedness of π_β gives the Carleson inequality and hence $C(\beta) \leq \|\pi_\beta\|$.

It remains only to compare the two indexing conventions. The $k > n$ sum is contained in the $k \geq n$ sum. In the other direction, on every parent d -adic atom, $d_n \beta$ is constant on each of the d children and has mean zero. Therefore its largest child value squared is at most d times its parent average:

$$|d_n \beta|^2 \leq d \mathbb{E}_{n-1} |d_n \beta|^2.$$

Taking the appropriate conditional sums yields the asserted comparison with a constant depending only on d . Tensoring a bounded scalar operator with an identity preserves its norm, which proves the Hilbert-valued statement. $\square$

Proof of Theorem 1. Put $H = L_2(\mathcal{M}, \tau)$, and represent $\mathcal{A}$ faithfully on H by left multiplication $a \mapsto L_a$. This is an abelian von Neumann algebra of normal operators. By the spectral representation theorem for abelian von Neumann algebras [2, Ch. 9], there are a measure space (Ω, ν) , a measurable field of Hilbert spaces (H_ω) , and a unitary

$$U : H \longrightarrow \int_\Omega^\oplus H_\omega d\nu(\omega)$$

such that every L_a , $a \in \mathcal{A}$, acts in the fibers as multiplication by a scalar function $\widehat{a}(\omega)$.

Apply this representation to the strongly measurable symbol b . After discarding a null set, write

$$UL_{b(t)}U^{-1}|_{H_\omega} = \beta_\omega(t)I_{H_\omega}.$$

Time conditional expectations commute with this fixed fiber decomposition; thus $d_k b$ corresponds to $d_k \beta_\omega$.

For integers $r \leq s$, let

$$\pi_b^{[r,s]} f = \sum_{k=r}^s d_k b \mathbb{E}_{k-1} f.$$

Conjugating by $I_{L_2(\mathbb{R})} \otimes U$ gives the decomposable operator

$$\int_\Omega^\oplus (\pi_{\beta_\omega}^{[r,s]} \otimes I_{H_\omega}) d\nu(\omega).$$

Consequently,

$$\|\pi_b^{[r,s]}\| = \operatorname{ess\,sup}_{\omega \in \Omega} \|\pi_{\beta_\omega}^{[r,s]}\|. \quad (3)$$

The positive operator $\mathbb{E}_n \sum_{k \geq n} |d_k b|^2$ lies in $\mathcal{A}$ and has fiber value $\mathbb{E}_n \sum_{k \geq n} |d_k \beta_\omega|^2$. The norm of an element of an abelian von Neumann algebra is the essential supremum of its fiber values. Since the levels are countable, the two suprema can be interchanged, giving

$$\|b\|_{\operatorname{BMO}_{c,d}} = \operatorname{ess\,sup}_{\omega \in \Omega} \|\beta_\omega\|_{\operatorname{BMO}_d}. \quad (4)$$

If the right side of (4) is finite, Lemma 3 and (3) give a bound, independent of r, s ,

$$\|\pi_b^{[r,s]}\| \leq C_d \|b\|_{\operatorname{BMO}_{c,d}}.$$

On finite martingales the truncations stabilize. Uniform boundedness therefore defines a bounded operator π_b on the dense subspace of finite martingales, and then on all of L_2 , proving the upper estimate in (2).

Conversely, if π_b is bounded, every finite interval truncation is bounded by $\|\pi_b\|$ (equivalently, one may project its output onto the corresponding orthogonal martingale-difference levels). Equations (3)–(4), the lower scalar estimate, and monotone passage through finite level intervals yield

$$\|b\|_{\operatorname{BMO}_{c,d}} \leq c_d^{-1} \|\pi_b\|.$$

This proves the theorem. □

5 Verification notes and limitations

- The proof is dimension-free because diagonalization produces scalar fibers before the scalar Carleson estimate is applied.
- The ambient algebra $\mathcal{M}$ need not be abelian. Inputs remain $L_2(\mathcal{M})$ -valued; the abelian hypothesis concerns only the von Neumann algebra generated by the symbol range.
- The proof uses finite level truncations, so no unjustified pointwise convergence of the infinite paraproduct series is required.
- The theorem does not address noncommuting symbols. In that regime the fiber reduction fails, and the logarithmic matrix obstruction cited by Wei and Zhang remains relevant.
- The most useful human check is the measurable direct-integral step for the desired level of generality (especially nonseparable ambient algebras). For the standard separable-predual setting, or for essentially separably valued symbols, the displayed decomposition is routine. The same proof can be phrased inside the abelian von Neumann tensor product to avoid separability.

Novelty check. The run’s lightweight indexes and bounded arXiv searches performed on 19 July 2026 for the phrases “operator-valued martingale paraproduct commuting symbol”, “semicommutative paraproduct abelian”, and “commuting normal operator symbol BMO” found no explicit statement of Theorem 1. The result may be folklore implicit in the abelian spectral theorem and scalar Carleson embedding, so this packet claims a likely-valid partial result, not a high-confidence literature novelty claim.

References

- [1] Z. Wei and H. Zhang, *Boundedness of operator-valued commutators involving martingale para-products*, arXiv:2401.08729 (2024).
- [2] R. V. Kadison and J. R. Ringrose, *Fundamentals of the Theory of Operator Algebras, Vol. II*, Academic Press, 1986.
- [3] F. Nazarov, G. Pisier, S. Treil, and A. Volberg, *Sharp estimates in vector Carleson imbedding theorem and for vector paraproducts*, J. Reine Angew. Math. 542 (2002), 147–171.